

Understanding CI/CD

Continuous integration and continuous delivery/deployment are intrinsic to DevOps. Here's a brief overview of how these practices help in the software development cycle.

CI/CD is a popular term for continuous integration (CI) and continuous delivery (CD). It is a set of practices that automate the software development process, and aim to streamline and accelerate software delivery from development to production effectively and faster.

Let us demystify a few terms.

Continuous integration (CI): CI frequently merges code changes into a remote or shared repository such as GitLab, GitHub, or Bitbucket. Every code change in the remote repository triggers an automated build and test jobs. This helps to find issues early in

the software development cycle and hence to recover faster from them.

Continuous delivery (CD): Continuous delivery is the next step after CI and enables automation in non-production environments. It makes sure that the application is always ready for deployment.

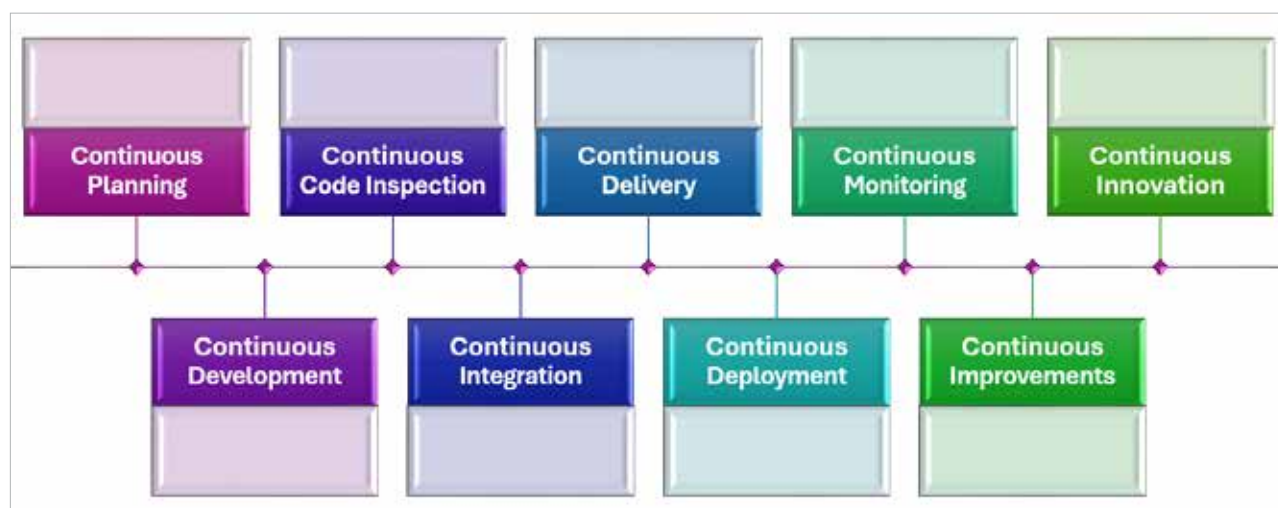


Figure 1: Continuous integration (CI) and continuous delivery (CD)



Figure 2: Key benefits of CI/CD

Continuous deployment (CD):

Continuous deployment is a more automated approach. It helps to automate the delivery of code changes to a production environment.

Benefits of CI/CD

CI and CD provide multiple advantages for project teams in most organisations (Figure 2).

By automating the code quality check, build, test, and deployment processes, CI/CD pipelines help to reduce the time it takes to deliver new features and bug fixes. They help to fail and recover faster with less spent in terms of money and time. Human intervention in the application lifecycle